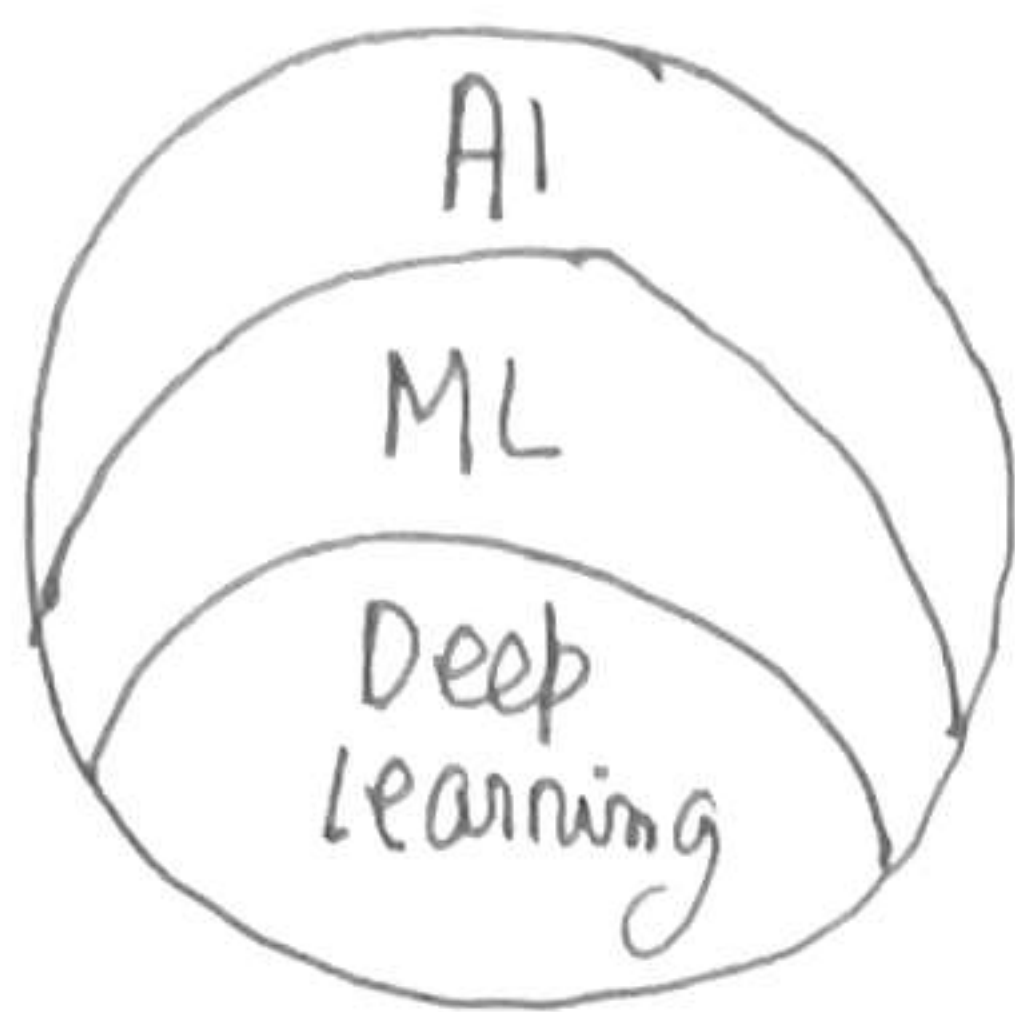


Advanced ML



Types of ML -

- (i) Supervised — Classification, Regression
- (ii) Unsupervised — Clustering
- (iii) Semi Reinforcement

1) Supervised Learning -

- Trained on labeled data (input + correct output)
- Learn a mapping from input to output
- Used to predict unseen data
- Eg - House price prediction
- Algorithms - Linear regression, Decision Trees, SVM

2) Unsupervised Learning -

- Trained on unlabeled data
- Finds patterns, structures, or clusters in data
- No predefined output labels
- Eg - Customer segmentation
- Algorithms - K-means, DBSCAN, PCA

3) Reinforcement Learning -

- Agent interacts with environment
- Learns via rewards & penalties
- Goal: maximize cumulative reward
- No labeled data or fixed instructions
- Eg - Robot navigation
- Algorithms - Q-Learning, ~~PDP~~ DQN

Additional types -

4) Semi supervised

= use few labeled + many

unlabeled data for training

5) Self supervised

= model creates its own labels from input data

6) Generative AI

= generates new content similar to training data

Regression -

- Supervised learning algorithm
- Used to predict continuous numerical values
- Goal is to find a function that maps the input features to a real valued output
- This function should minimize the difference between predicted values and the actual values in the training data.

- Types:

- Linear Regression — Simple
- Polynomial — Multiple
- ~~Additional~~ Additionally — Ridge
- — Lasso
- — Elastic Net

Linear Regression -

- Predicts a continuous value using a linear relationship
- Models relation between dependent (Y) and independent (X) variable
- Supervised learning algorithm
- Fits a best fit straight line (regression line) to data
- Used for prediction and trend analysis
- Regression line = line showing relationship between X and Y
- Relationship Types:

└ Positive = $X \uparrow Y \uparrow$
└ Negative = $X \uparrow Y \downarrow$

where, X = Independent Variable (Input / Predictor)
Y = Dependent Variable (Output / Target)

- Types:

└ Simple = One independent variable
└ Multiple = More than one independent variable

- Goal = Minimize error & find the best fitting straight line for accurate prediction

- Linear equation: $Y = w_0 + w_1 X + \epsilon$ where, Y = dependent variable (target)
X = independent variable (feature)

- Multiple Linear equation: $Y = w_0 + w_1 X_1 + w_2 X_2 + \dots + w_p X_p + \epsilon$
where, $X_1, X_2, \dots, X_p$ = independent variables
 $w_0, w_1, \dots, w_p$ = coefficients for these variables
 ϵ = error term

w_0 = y intercept of the line
 w_1 = slope of the line representing effect of X on Y

ϵ = error term

- Error is measured using a cost (loss) function
- Model parameters are updated to minimize cost
- Hypothesis = Assume linear relation between X and Y
- Cost Function = Measures difference between actual & predicted values
- Optimization = Minimize cost by updating parameters (weights)
- Hypothesis Function:

└ Simple $\Rightarrow \hat{y} = w_0 + w_1 X$

└ Multiple $\Rightarrow \hat{y} = w_0 + w_1 X_1 + w_2 X_2 + \dots + w_p X_p$

- Loss Function:

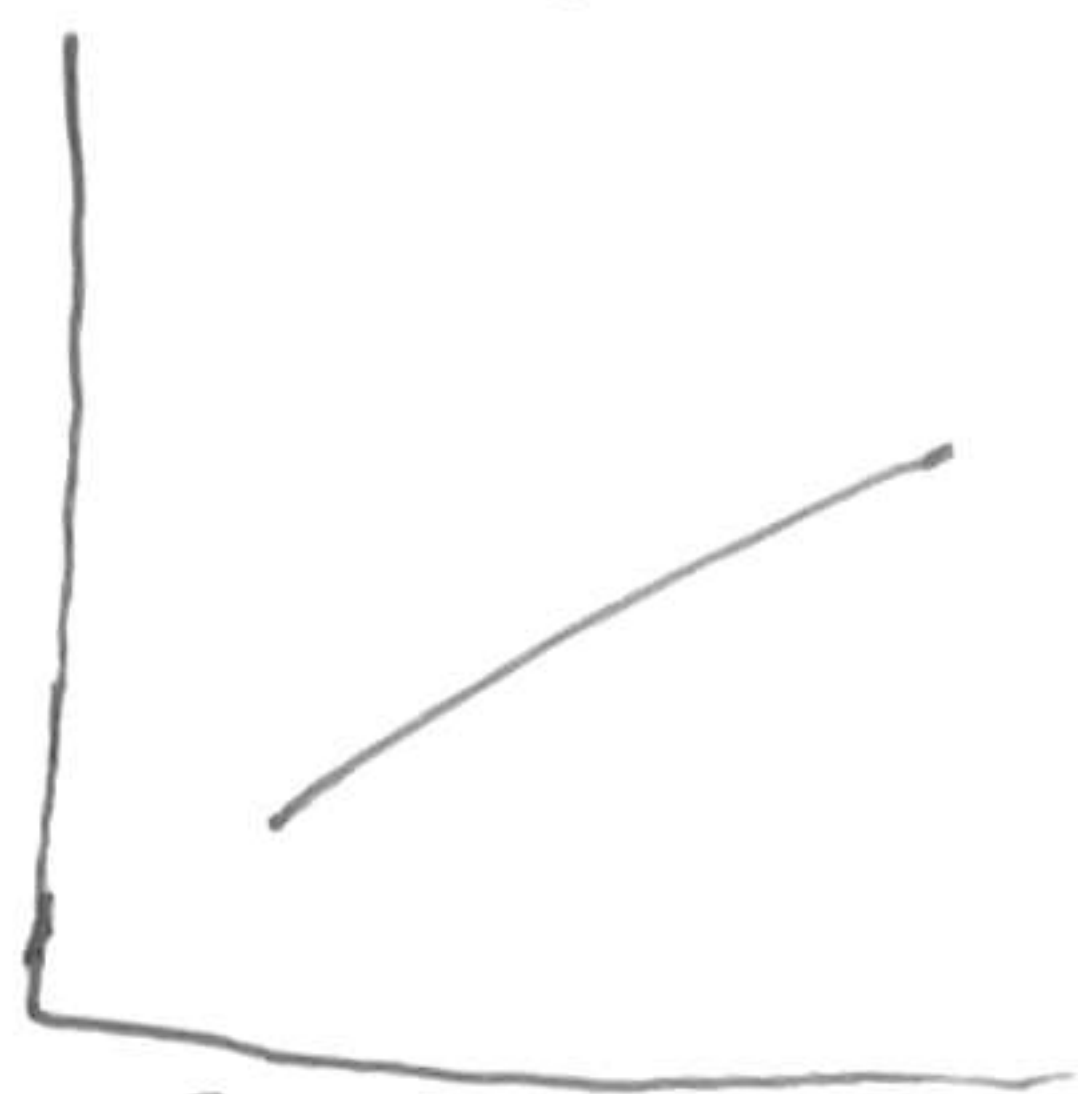
$$J(w_0, w_1) = \frac{1}{2n} \sum (Y_i - \hat{Y}_i)^2$$

Y_i = observed value

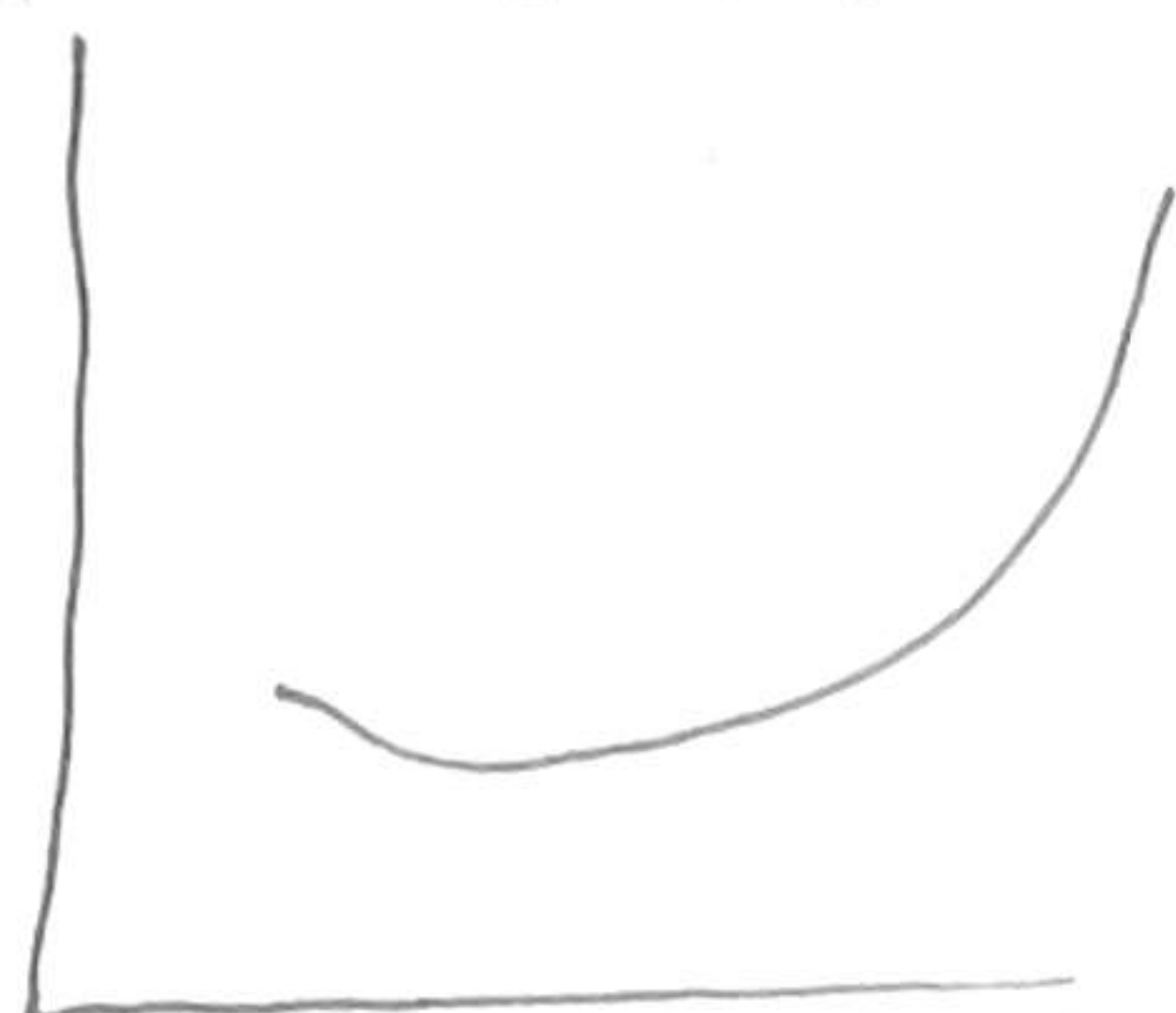
$\hat{Y}_i$ = predicted value

Polynomial Regression -

- Relationship between the independent variable and dependent variable is modeled as an n th degree polynomial function



Simple Linear Model



Polynomial Model

- Equation: $y = w_0 + w_1x + w_2x^2 + w_3x^3 + \dots + w_nx^n + \epsilon$

- Need for Polynomial Regression:

- Non linear relationship
- Better fit for curved data
- Flexibility & Complexity

- Advantages:

- Interpretability
- Speed
- Predictive analytics
- Linear Relationships
- Simplicity
- Efficiency

- Common Challenges:

- Overfitting
- Multicollinearity
- Outliers and their impact

- Applications:

- Predictive modeling
- Feature selection
- Financial Forecasting
- Risk management

Maxima & Minima -

Types

Local Maxima (nearly)
Global Maxima (entire domain)

Types

Local Minima (nearly)
Global Minima (entire domain)

Mathematical equation:

(i) Local minima:

$$\text{if } f(x_0) < f(x) \text{ for all } x$$

(ii) Local maxima:

$$\text{if } f(x_0) > f(x) \text{ for all } x$$

How we find maxima & minima?

(i) Find derivative test

(ii) Second derivative test

(iii) $f'(x)$

If $f'(x)$ changes from + to - $\Rightarrow$ Local minima

If $f'(x)$ changes from - to + $\Rightarrow$ Local maxima

No sign change $\Rightarrow$ Neither

(iv) Second derivative

$$f''(x)$$

$$Q. f(x) = x^3 - 3x^2 + 2$$

First derivative

$$f'(x) = 3x^2 - 6x$$

$$3x^2 - 6x = 0$$

$$3x(x-2) = 0$$

$$x = 0, 2$$

Interval

$$(-\infty, 0)$$

$$(0, 2)$$

$$(2, +\infty)$$

Interval

Interval

$$x < 0$$

$$0 < x < 2$$

$$x > 2$$

Sign

+

-

$$1) 3x(x-2)$$

$$x = -1$$

$$= 3(-1)(-3)$$

$$= 9$$

$$2) 0 < x < 2$$

$$x = 1$$

$$f'(1) = 3(1)(1-2)$$

$$= -3$$

$$3) x > 2$$

$$x = 3$$

$$f'(3) = 3(3)(1)$$

$$= 9$$

~~Second derivative~~

Second derivative -

$$f''(x)$$

Local maxima

(i) If $f'(c)=0$ and $f''(c)<0$
curve is concave

Local minima
(ii) If $f'(c)=0$ and $f''(c)>0$
curve is convex

Rank in matrix -

Rank = max no. of linearly independent rows & columns in matrix

(i) Determinant method -

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}$$

$$\begin{aligned} \det(B) &= 1(4-6) - 2(2-6) + 3(2-4) \\ &= -2 + 8 - 6 \\ &= 0 \end{aligned}$$

$$\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0 \quad \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0$$

$$\begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix} = 2 - 4 = -2 \neq 0$$

$$\text{Rank} = 2$$

(ii) Echelon form -

$$\begin{bmatrix} 1 & _ & _ & _ & _ & _ \\ 0 & 1 & _ & _ & _ & _ \\ 0 & 0 & 1 & _ & _ & _ \\ 0 & 0 & 0 & 1 & _ & _ \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & -1 & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4 \quad R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 6 & 3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 6 & 3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_1 \rightarrow R_1 - R_4 \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4$$

$$R_3 \leftrightarrow R_4$$

$$R_2$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 2$$

Maxima & Minima of function of several variables -

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

Input $\rightarrow n$ real numbers
Output $\rightarrow$ one real number

general form, $f = f(x_1, x_2, x_3, \dots, x_n)$
 $\uparrow$ dependent $\rightarrow$ independent

eg- $f(x, y) = x^2 + y^2$
 $\uparrow$ input $\rightarrow$ output

$$f(1, 2) = 1^2 + 2^2 = 5$$

Types:

- (i) Local Maxima
- (ii) Local Minima
- (iii) Saddle point (neither max or min)

Convexity of Error function -

Convex function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if for all $x_1, x_2 \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, the following condition holds true

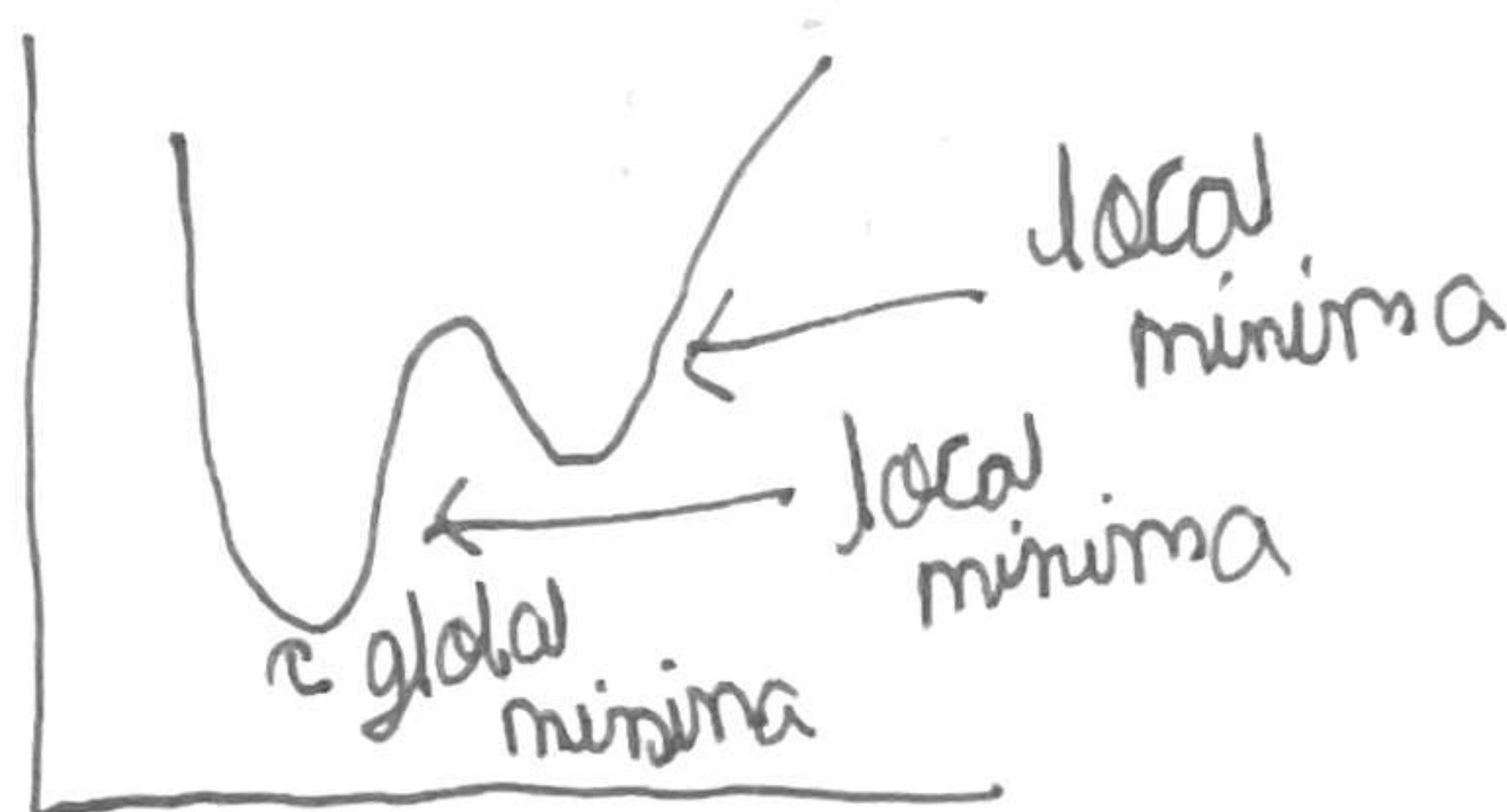
$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

$x_1, x_2 \rightarrow 2$ input points



Concave function: Downwards curve

$$f(\lambda x_1 + (1-\lambda)x_2) \geq \lambda f(x_1) + (1-\lambda)f(x_2)$$



Loss / cost

minimize $\theta, \theta_1 \frac{1}{m} \sum_{i=1}^m (h(x^i) - y^i)^2$

$$\left. \begin{array}{l} x^i = \text{predicted} \\ y^i = \text{actual} \end{array} \right\} \text{ } i^{\text{th}} \text{ training data}$$

$m = \text{no. of training example}$

difference between actual and predicted value

Regression model by solving normal equation

Linear regression: best fit line

Simple: $y = mx + b$

multiple eqn $\Rightarrow y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$

Normal eqn:

$$\theta = (X^T X)^{-1} X^T Y$$

$X \rightarrow$ input function / feature

$Y \rightarrow$ output feature value

$\theta \rightarrow$ hypothesis / parameter / coefficient

$(X^T X)^{-1} \rightarrow$ Inverse

Gradient Descent -

~~$$\theta = \theta - \alpha * \nabla J\theta$$~~

1) ~~Entire data set training~~

~~$$\theta = \theta - \alpha \frac{1}{n} \sum_{i=1}^n \nabla J_i(\theta)$$~~

2)

$\rightarrow$ How it works?

(i) Weight & bias

(ii) Loss function (error)

(iii) Minimize the loss

$$\theta = \theta - \alpha * \nabla J\theta$$

Types

$\rightarrow$ Batch gradient

$\rightarrow$ Stochastic

$\rightarrow$ Mini Batch

1) Entire dataset training :

$$\theta = \theta - \alpha \frac{1}{n} \sum \nabla J_i(\theta)$$

2) Only one training (single sample) :

$$\theta = \theta - \alpha J_i(\theta)$$

3) Small batch :

$$\theta = \theta - \alpha \frac{1}{n} \sum \nabla J_i \theta$$

Gradient Descent in Linear Regression -

(i) $y = mx + b$

(ii) Cost $\sum_{i=1}^n (MSE)$

$$J(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

(iii) GD updated rules :

$$\text{weight } (w) \quad \frac{\partial J}{\partial w} = -\frac{2}{n} \sum_{i=1}^n x_i (y_i - \hat{y}_i)$$

$$\text{Bias } (b) \quad \frac{\partial J}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

(iv) Core gradient descent

$$\text{weight } w = w - \alpha \frac{\partial J}{\partial w}$$

$$\text{bias } b = b - \alpha \frac{\partial J}{\partial b}$$

$$w_n = w_0 - \eta \frac{\partial J(\theta)}{\partial w}$$

$$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

$$y - \hat{y}$$

data 100x2

data [1 → 100, 1 → 2]

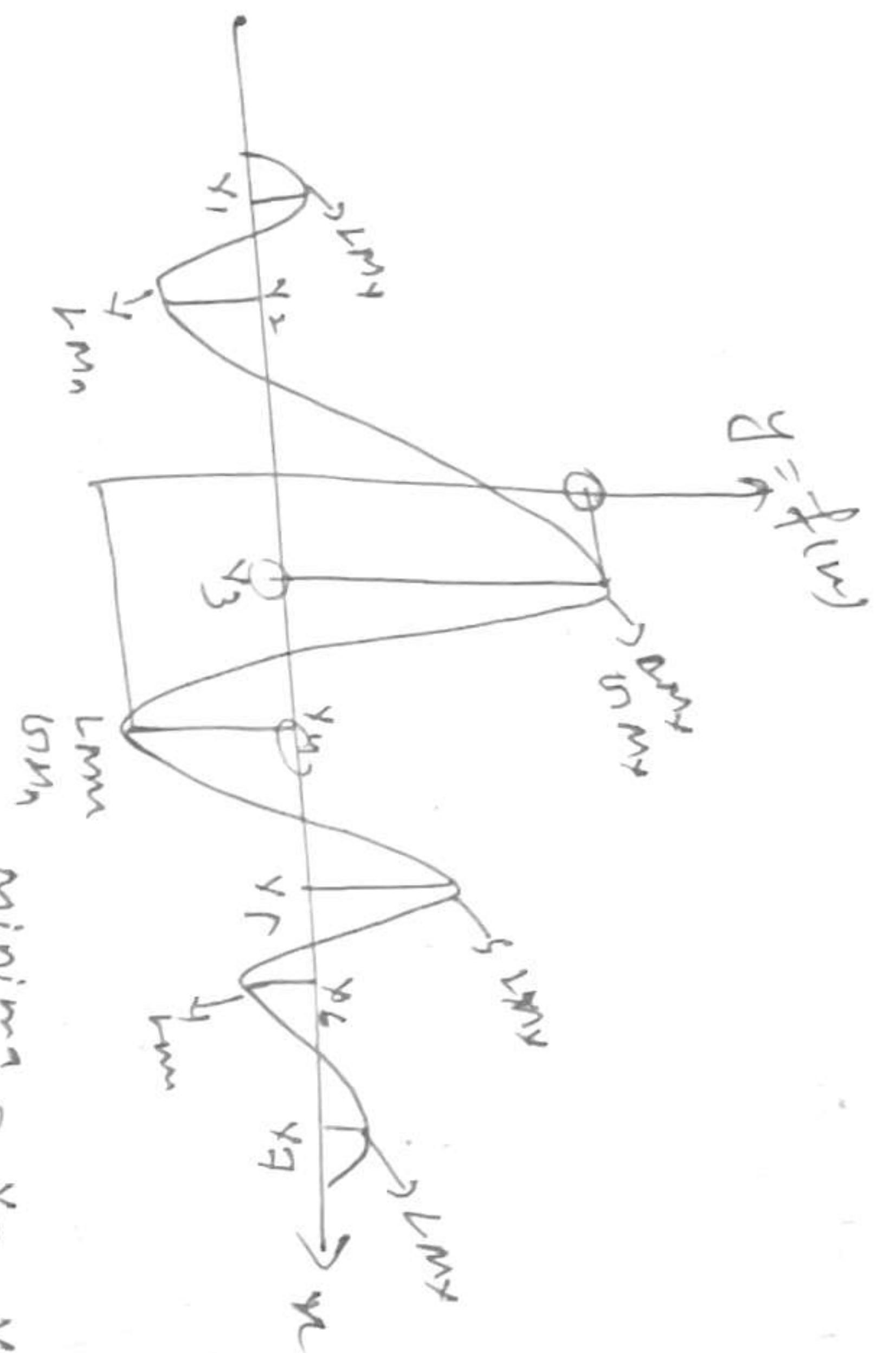
data [1] [2] [3]

data [1-100] [11-20] in

Δ w/b

$\frac{\partial L}{\partial w/b}$

$J(\theta)$
loss



Minima = x_2, x_4, x_6
 Maxima = x_1, x_3, x_5, x_7 .

Minima points

Local value

Value of x and y

Rank in a Matrix —

- If matrix is of the order 3×3 , then the max rank of a matrix can be 3.

- Nullity: No. of vectors in the null space

(OR) dimension of the null space of the given matrix

- Total columns = Rank + Nullity

- Definition: max number of linearly independent rows or columns in a matrix

- Properties:

(i) $P(A)$ = order of A if A is a non-singular matrix.

(ii) $P(A)$ = no. of non zero rows if it is in Echelon form

(iii) $P(A)$ = order of identity matrix if it is in Normal form

(iv) $P(A)$ < order of A if it is a singular matrix

- Use:

(i) Checking linear independence of features

(ii) Uniqueness of regression solution

(iii) Detecting multicollinearity

(iv) Data quality & redundancy checked

Positive & Negative Definite / Semi Definite Matrices —

- Quadratic form: $Q(x) = x^T A x, x \neq 0$

(i) Positive definite matrix:

if $x^T A x > 0 \forall x \neq 0$

Properties:

- All eigenvalues $\lambda_i > 0$

- A must be symmetric $A = A^T$

- Determinants of leading principle minors > 0

- Quadratic form always yields a positive value

Eg. $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow x^T A x = 2x_1^2 + 3x_2^2 > 0$

(ii) Positive Semi definite matrix:

if $x^T A x \geq 0 \forall x \neq 0$

Properties:

- Eigenvalues $\lambda_i \geq 0$

- Symmetric in nature

- Quadratic form never negative, but may be zero

Eg. $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow x^T A x = x_1^2 \geq 0$

(iii) Negative Definite Matrix:

$$\text{if } x^T A x < 0 \quad \forall x \neq 0$$

Properties

- All eigenvalues $\lambda_i < 0$
- Symmetric matrix
- Quadratic form always yields a negative value

eg- $A = \begin{bmatrix} -4 & 0 \\ 0 & -2 \end{bmatrix} \Rightarrow x^T A x = -4x_1^2 - 2x_2^2 < 0$

(iv) Negative Semi-definite Matrix:

$$\text{if } x^T A x \leq 0 \quad \forall x \neq 0$$

Properties

- Eigenvalues $\lambda_i \leq 0$
- Always non positive
- May become zero for some non zero vectors

eg- $A = \begin{bmatrix} -3 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow x^T A x = -3x_1^2 \leq 0$

* Summary -

Type	Condition on $x^T A x$	Eigenvalue condition
Positive definite	> 0	all $\lambda > 0$
Positive semi definite	≥ 0	all $\lambda \geq 0$
Negative definite	< 0	all $\lambda < 0$
Negative semi definite	≤ 0	all $\lambda \leq 0$

Eigen values -

Unique scalar values linked to a matrix or linear transformation

Eigen vectors -

Non zero vectors that, when multiplied by a matrix, only ~~stretch~~ stretch or shrink without changing direction

Q. Find the eigen values & eigen vectors of the following matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & 6 \\ 0 & 0 & -3 \end{bmatrix}$$

And Characteristics

matrix:

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & -2-\lambda & 6 \\ 0 & 0 & -3-\lambda \end{vmatrix} = 0$$

Characteristic Polynomial: $\Rightarrow \lambda^3 - p\lambda^2 + Q\lambda - |A| = 0$

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & -2 & 6 \\ 0 & 0 & -3 \end{vmatrix} \begin{array}{l} \text{Sum of} \\ \text{these elements (diagonals)} \\ p = -4 \end{array} \quad \begin{array}{l} Q = \text{sum of minors of diagonals} \\ = \begin{vmatrix} -2 & 6 \\ 0 & -3 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 0 & -3 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} \\ = 6 - 3 - 2 = 1 \end{array}$$

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -2 & 6 \\ 0 & 0 & -3 \end{vmatrix} = 1(6-0) - 2(0) + 3(0) = 6$$

$$\boxed{\lambda^3 + 4\lambda^2 + \lambda - 6 = 0}$$

Solutions, $\lambda = 1, -2, -3$ (Eigen values)

Now, find out the eigen vectors:
using the given eqn

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & -2-\lambda & 6 \\ 0 & 0 & -3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{for } \lambda = 1, \begin{bmatrix} 0 & 2 & 3 \\ 0 & -3 & 6 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$0x_1 + 2x_2 + 3x_3 = 0 \quad \text{--- (a)}$$

$$0x_1 - 3x_2 + 6x_3 = 0 \quad \text{--- (b)}$$

$$0x_1 + 0x_2 - 4x_3 = 0 \quad \text{--- (c)}$$

from (a) & (b) using Cramers Rule

$$\frac{x_1}{\begin{vmatrix} 2 & 3 \\ -3 & 6 \end{vmatrix}} = -\frac{x_2}{\begin{vmatrix} 0 & 3 \\ 0 & 6 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 0 & 2 \\ 0 & -3 \end{vmatrix}}$$

$$\frac{x_1}{21} = -\frac{x_2}{0} = \frac{x_3}{0}$$

divide by 21

$$\frac{x_1}{1} = -\frac{x_2}{0} = \frac{x_3}{0}$$

$$\text{So, } x_1 = 1$$

$$x_2 = 0$$

$$x_3 = 0$$

$$\text{(OR)} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Maxima & Minima -

Q. Find the maximum & minimum values of

$$y = x^3 - 9x^2 + 24x \quad \text{--- (1)}$$

Ans 1) diff y wrt x

$$\frac{dy}{dx} = 3x^2 - 18x + 24 \quad \text{--- (2)}$$

$$2) \frac{dy}{dx} = 0$$

$$3x^2 - 18x + 24 = 0$$

$$\Rightarrow x^2 - 6x + 8 = 0$$

$$(x-2)(x-4) = 0$$

$$x = 2, 4 \rightarrow \text{Critical values}$$

3) diff (2) wrt x

$$\frac{d^2y}{dx^2} = 6x - 18 \quad \text{--- (3)}$$

Now put the values of x in (3)

for $x = 2$

$$\left[\frac{d^2y}{dx^2} \right]_{x=2} = (6 \times 2) - 18 = -6 \text{ (-ve)}$$

This value is -ve, means $-6 < 0$

This is maxima

$f(x)$ is maximum at $x = 2$

Now, to find out max value of $f(x)$

put $x = 2$ in (1)

$$y = 2^3 - 9(2)^2 + (24 \times 2)$$

$$= 8 - 36 + 48$$

$$= 20 \text{ (This is max value at } x = 2)$$

for $x = 4$

$$\left[\frac{d^2y}{dx^2} \right]_{x=4} = (6 \times 4) - 18 = 6 \text{ (+ve)}$$

This value is +ve, means $6 > 0$

This is minima

$f(x)$ is minimum at $x = 4$

Now, to find out minimum value of $f(x)$

put $x = 4$ in (1)

$$y = 4^3 - 9(16) + (24 \times 4)$$

$$= 16 \text{ (This is the min value at } x = 4)$$

Gradient Descent -

- Algorithms to perform optimization
- Way to minimize an objective function $J(\theta)$ parameterized by a model's parameter: $J(\theta) = \text{loss}(w, b)$
- Learning rate η determines the size of the steps we take to reach a (local) minimum
- We follow the direction of the slope of the surface created by the objective function downhill until we reach a valley

epochs = 5

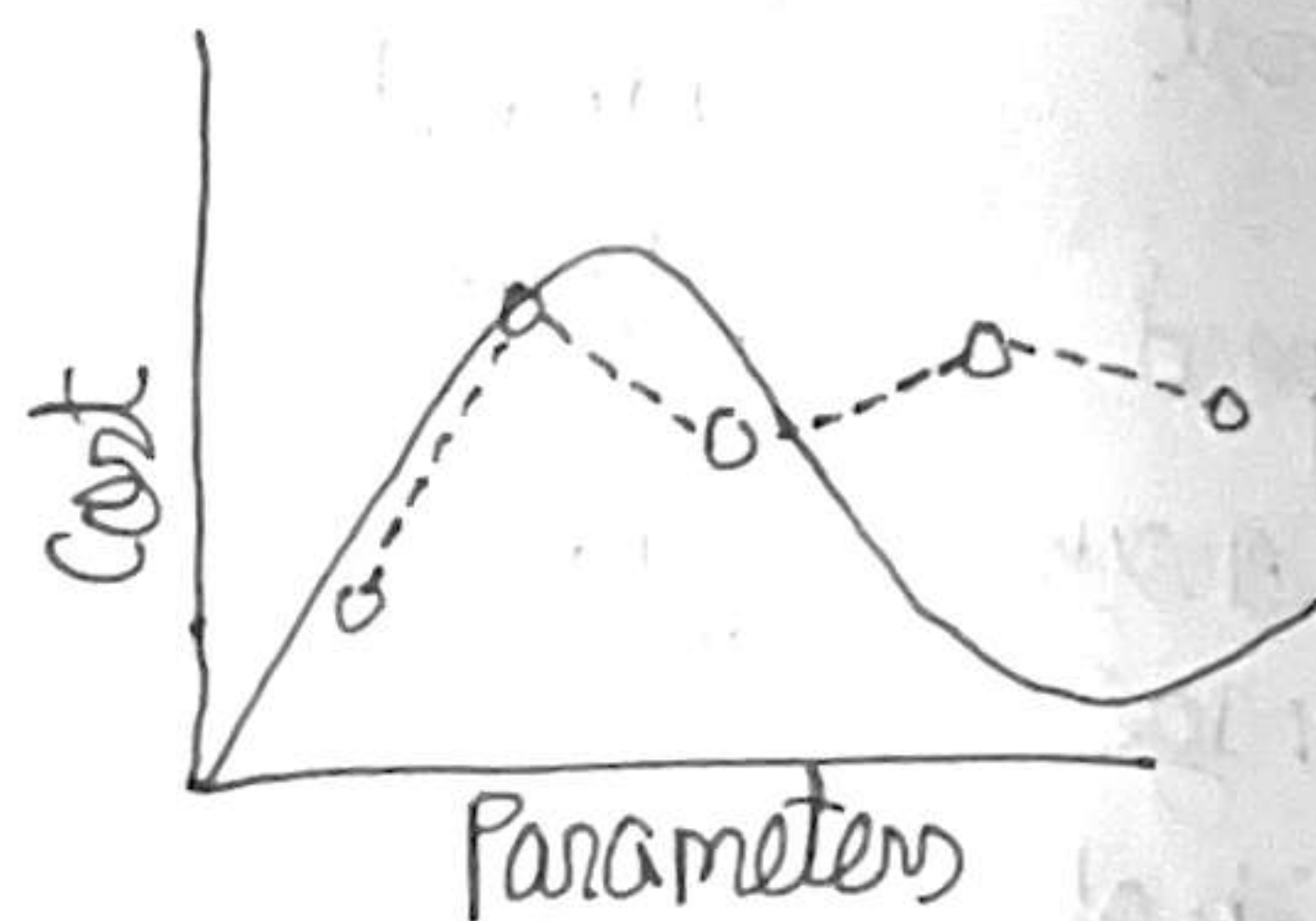
for i in range(epochs):

for j in range(data.shape[0]):

- select 1 row of data
- predict (using forward propagation)
- calculate loss using loss function
- update weights and bias using GD

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w_{\text{old}}}$$

→ calculate avg loss of the epochs



Types of Gradient Descent -

- 1) Batch Gradient or Vanilla Gradient (Default)
- 2) Stochastic Gradient
- 3) Mini-batch Gradient

Advantages -

- (i) Easy to use
- (ii) Fast updates
- (iii) Memory efficient
- (iv) Usually finds a good spot

Disadvantages -

- (i) Slow for giant hills (large datasets)
- (ii) Can get stuck in shallow dips (low minima)
- (iii) Finding the perfect push (learning rate)

Challenges -

- (i) Overfitting
- (ii) Learning rate selection
- (iii) Local optima
- (iv) Convergence rate
- (v) Saddle points

- Minimizes the MSE which serves as the loss function

- For weight: $w = w - \alpha \frac{\partial J(w, b)}{\partial w}$ | - For bias: $b = b - \alpha \frac{\partial J(w, b)}{\partial b}$

Applications -

- (i) Training regression models by minimizing loss functions
- (ii) Optimizing parameters of neural networks & deep learning models
- (iii) Training deep learning models on image, text & speech data
- (iv) Efficient use of GPUs/TPUs

Types of Gradient Descent —

Aspect	Batch	Stochastic	Mini Batch
1) Data used per update	Entire data training dataset	Single training example	Small subset of data (batch)
2) Update frequency	One time update per epoch	One update per sample	Multiple updates per epoch
3) Computation cost	Very high for large datasets	Very low per update	Moderate
4) Speed	Slow	Fast but unstable	Fast & stable
5) Memory usage	High	Very low	Moderate
6) Risk of local minima	High	Low	Low
7) Best suited for	Small datasets	Very large / streaming data	Most real world ML tasks
8) Common usage	Rare in practice	Online learning	Deep Learning

* Random Variables: It is a set of possible values from a random experiment

Random variable is denoted by capital letters

$$X = \{1, 2, 3, 4, 5, 6\}$$

↳ Sample Space

Types of Random Variables:

- (i) Discrete Random Variable
- (ii) Continuous Random Variable

Probability distributions:

It is a list of all the possible outcomes of a random variable along with their corresponding probability values.

eg - 1) A random experiment: Coin toss

Coin Toss	1 (H)	0 (T)
Probability	$1/2$	$1/2$

2) 1 Dice Roll

$y=f(x)$	x (Dice Roll)	1	2	3	4	5	6
	y (Probability)	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

3) 2 Dice Roll

	(X,Y)	1	2	3	4	5	6
	1	2	3	4	5	...	
Sum of	2	:	:	:	:	:	:
2 die	3	:	:	:	:	:	:
	4	:	:	:	:	:	:
	5	:	:	:	:	:	:
	6	:	:	:	:	:	:

Probability Distribution Table

$$P(2) = 1/36$$

$$P(3) = 2/36$$

$$P(4) = 3/36$$

$$P(12) = 1/36$$

Probability distribution function: a math ~~func~~ relation between all the possible outcomes of RV with possible probability

3) Poisson -

Gives the no. of events occurred at a specific time ~~be~~ period with the help of its mean

$$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Types of Probability Distributions:

(i) Discrete

(ii) Continuous

- There are some popular probability distributions:

- Normal
- t-distribution
- Uniform
- Bernoulli (discrete)
- Gamma
- Binomial (discrete)
- Poisson (discrete)
- Geometric (discrete)

1) Bernoulli

Probability of success p if the value of random ~~value~~ variable is 1 & probability of failure $1-p$ if the random variable is 0.

$$P(X=x) = \begin{cases} p, & x=1 \\ 1-p, & x=0 \end{cases}$$

2) Binomial

Includes no. of trials n , probability of success & probability of failure

$$P(X=x) = {}^nC_x p^x (1-p)^{n-x}$$

Normal Distribution -

- Also known as Gaussian distribution
- Commonly used in statistical analysis that is symmetrical around the mean with a bell shaped curve.
- A continuous probability distribution
- Characterized by 2 parameters:
 - (i) Mean (μ)
 - (ii) Standard deviation (σ)
- Represents the centre of distribution while the standard deviation represents the spread of the distribution
- Denoted as: $X \sim N(\mu, \sigma)$ [X belongs to normal distribution]

Random Variable

- Importance:

- (i) Many natural phenomena follow a normal distribution
- eg - Height, weight of people
- (ii) PDF equation of Normal Distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\Rightarrow$ If $\mu=0$ and $\sigma=1$, then it is known as Standard Normal Variate (Z)

$$Z \sim N(0, 1)$$

Student's T-distribution -

Condition:

- (i) Sample size (n) < 30 (T-distribution)
- (ii) Sample size (n) ≥ 30 (Normal Distribution)
- Population is normally distributed
- Population standard deviation is unknown
- PDF is calculated using:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$
- Range from $-\infty$ to $+\infty$

constant over under curve = 1

$$x = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

X will be $-\infty < x < +\infty$
 $\uparrow$ sample $\uparrow$ (T $\rightarrow$ Normal)

- Variance always greater than 1 & is limited only to 3 or more dof.
- When to use:
 - Sample size is 30 or less than 30 & SD is unknown
 - population distribution must be unimodal & skewed

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Def: Used to make presumptions about a mean when SD is not known.

Probability distribution function -

A mathematical function that describes the probability of obtaining different values of a random variable in a particular probability distribution.

(a) Probability Mass Function (PMF):

If you are defining the function for Discrete Random Variable (Coin Toss)

(b) Probability Density Function (PDF):

If you are defining the function for Continuous Random Variable

(c) Cumulative Distribution Function (CDF):

It is the combination of both PMF & PDF

1) Probability Mass Function (PMF) -

- Mathematical function that describes the probability distribution of a discrete random variable.

- 2 conditions:

- (i) The probability assigned to each value must be non negative (i.e. greater than or equal to zero)
- (ii) The sum of the probabilities assigned to all possible values must equal to 1.

2) Cumulative Distribution Function (CDF) -

Describes the probability that a random variable X with a given probability distribution will be found at a value less than or equal to X

$$P(X \leq x) = P(X \leq x)$$

Describes the probability that a random variable X with a given probability distribution will be found at a value less than or equal to X

$$P(X=3) = 1/6 \quad P(X=4) = 1/6 \quad \dots$$

$$\ln \text{ CDF, } F(X \leq 4) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6}$$

$$P(X=4) \quad P(X=2) \quad P(X=1)$$

3) Probability Density Function (PDF) -

- Mathematical function that describes the probability distribution of a continuous random variable
- Used to find out the probability of any data in a particular range.

Maxima & Minima of function of several variables
- It refers to the highest & lowest points of a function's graph

Q. Find maxima & minima of the function:

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20 \quad \text{--- (1)}$$

Ans

1) Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ (Partial derivatives)

$$\frac{\partial f}{\partial x} = 3x^2 - 3 \quad \text{and} \quad \frac{\partial f}{\partial y} = 3y^2 - 12$$

2) Compute $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y^2}$ (1 and 2)

$$\frac{\partial^2 f}{\partial x^2} = 6x = 1 \quad \text{and} \quad \frac{\partial^2 f}{\partial y^2} = 6y = 2$$

3) Compute $\frac{\partial^2 f}{\partial x \partial y} = S$ (derivative of $\frac{\partial f}{\partial y}$ wrt x)

$$\frac{\partial^2 f}{\partial x \partial y} = 0 \quad (\text{because in } 3y^2 - 12, x \text{ is not there})$$

$$S = \frac{\partial^2 f}{\partial x \partial y} = 0$$

4) for maxima and minima

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0 \quad (\text{to find out the roots})$$

$$3x^2 - 3 = 0 \quad \text{and} \quad 3y^2 - 12 = 0$$

$$3x^2 = 3 \quad \text{and} \quad 3y^2 = 12$$

$$x^2 = 1 \quad \text{and} \quad y^2 = 4$$

$$\boxed{x = \pm 1}$$

$$\text{and} \quad \boxed{y = \pm 2}$$

There are total 4 roots & there are $(1, 2)$ $(1, -2)$
 $(-1, 2)$ and $(-1, -2)$

5) Now, need to check maxima & minima using H-S²

$$H - S^2$$

$$6x \times 6y = 0$$

$$36xy$$

$$H - S^2 = 36xy$$

6) Put the value of roots in $t - s^2$ and in t

- * There are some conditions for maxima & minima
- (i) if $t - s^2 > 0$ and $t > 0 \rightarrow$ minima (for min value)
 - (ii) if $t - s^2 > 0$ and $t < 0 \rightarrow$ maxima (for max value)
 - (iii) if $t - s^2 < 0$ then no need to check the t and in this case, there are no minima or maxima

Case (i) $(x, y) = (1, 2)$
 $t - s^2 = 36xy$ and $t = 6x = 6 \times 1$
 $= 72 > 0$ $= 6 > 0$

$(1, 2)$ is a minimum point & to compute the minimum value, put the value of $(x, y) = (1, 2)$ in ①

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20$$
$$f(1, 2) = 1^3 + 2^3 - 3 \times 1 - 12 \times 2 + 20$$
$$= 2$$

Minimum value for $(1, 2) \rightarrow$ minima is 2

Case (ii) $(x, y) = (1, -2)$

$$t - s^2 = 36xy$$
$$= 36 \times 1 \times (-2)$$
$$= -72 < 0$$

NO need to check t now

$f(x, y)$ is neither maxima nor minima at $(1, -2)$
 $(1, -2)$ is called a saddle point

Case (iii) $(x, y) = (-1, 2)$

$$t - s^2 = 36xy = 36 \times (-1) \times (2) = -72 < 0$$

Same case

$\Rightarrow (-1, 2)$ is also a saddle point

Case (iv) $(x, y) = (-1, -2)$

$$t - s^2 = 36xy$$
$$= 36(-1) \times (-2) = 72 > 0$$

$$\text{Now, } t = 6x = 6(-1) = -6 < 0$$

$f(x, y)$ is maximum at $(-1, -2)$
 $(-1, -2)$ is a maxima point

Application of discrete probability distribution -

- (i) Randomized algorithms (eg. Quick sort)
- (ii) Packet loss modeling (Bernoulli/Binomial)
- (iii) Naive Bayes Classification (discrete feature modeling)

Importance of continuous probability distribution & (Applications) -

- (i) Modeling uncertainty
- (ii) Gaussian distribution
- (iii) Maximum Likelihood Estimation (MLE)
- (iv) Bayesian Inference
- (v) Generative models

Applications of continuous probability distribution -

Discrete vs Continuous distribution -

Discrete

- 1) Random variable can only take on a finite or countable no. of values
- 2) Probability function assigns probabilities to each possible outcome
- 3) Probability of any single outcome is non-zero
- 4) Cumulative distribution function is stepwise
- 5) Used to model events with discrete outcomes
- 6) eg - binomial distribution, Poisson distribution, geometric distribution

Continuous

- 1) Random variable can take on any value within a certain range or interval
- 2) Probability function assigns probabilities to each possible value within a range or interval
- 3) Probability of any single value is zero
- 4) Cumulative distribution function is continuous
- 5) Used to model events with continuous outcomes
- 6) eg - Normal distribution, exponential distribution, beta distribution

5) Chi Squared distribution -

- Continuous
- That arises in statistics, particularly in hypothesis testing
- Single parameter, often denoted as k (degree of freedom)
- Mean = k
- Variance = $2k$

$$f(x) = \frac{1}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})} \left(\frac{x}{2}\right)^{\frac{k}{2}-1} e^{-x/2}$$

Now, to find out the maximum value, put the value of $(x, y) = (-1, -2)$ in ①

$$\begin{aligned} f(x, y) &= x^3 + y^3 - 3x - 12y + 20 \\ &= (-1)^3 + (-2)^3 - 3 \times (-1) - 12(-2) + 20 \\ &= (-1) - 8 + 3 + 24 + 20 \\ &= 38 \end{aligned}$$

Maximum value for $(-1, -2) \rightarrow$ maximum is 38

Final results:

- Maximum point = $(-1, -2)$
- Maximum value = 38
- and
- Minimum point = $(1, 2)$
- Minimum value = 2

Some distributions -

~~Discrete probability distribution~~

- 1) Geometric distribution -
- Discrete probability distribution
- Includes the success failure probability until the success
- Probability is encountered

$$G(X=x) = (1-p)^{x-1} \cdot p$$

2) Normal distribution (Gaussian distribution)

- Continuous probability distribution
- Bell shaped, symmetrical basic continuous probability distribution
- 2 factors:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

3) Uniform distribution -

- Continuous
- All values within a specified range are equally likely to occur
- Parameters: lower bound (a) & upper bound (b)
- $y = \frac{a+b}{2}$ $\sigma^2 = \frac{(b-a)^2}{12}$

$$f(x) = \frac{1}{b-a}$$

4) Exponential distribution -

- Continuous
- Represents duration between occurrences in a Poisson process which occurs continuously & independently at a constant rate.
- Parameter: Rate Parameter (λ) (mean)
- $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

R^2 (Coefficient of determination) -

- Relative measure that indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.

- Calculated as 1 minus the ratio of RSS to TSS

$$R^2 = 1 - \frac{SS_{Res}}{SS_{Tot}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- Unit: unitless

- Interpretation: Higher R^2 value indicates a better model fit

$R^2 \approx 1$: model predicts the variability in response variable

$R^2 \approx 0$: model explains none of the variability in response variable

- Adjusted R^2 is a modified form of R^2 that penalizes adding unnecessary predictors in a regression model

$$R^2_{adj} = 1 - \frac{(1 - R^2)(n - 1)}{(n - p - 1)}$$

RSE vs R^2 -

Aspect

1) Objective

R^2
To determine how much in the strength of relationship between independent & dependent variables

2) Measurement Focus
Lies between 0 and 1

3) Purpose & Use
Used for comparison between 2 models

4) Interpretation
Higher the value more is the variability between dependent & independent variable

5) Relationship between the 2 metrics
No / can't be compared relationship just as one decreases the other one increases & vice versa

RSE

To find the average distance between actual values and the predicted values through which the regression line passes

Does not lie between 0 and 1

Helps to determine the precision of predictions

Lower the value better is the model

No / can't be compared relationship just as one decreases the other one increases & vice versa

Normal distribution characteristics -

(i) Symmetry

(ii) Mean = Median = Mode

(iii) Bell shaped curve

(iv) Standard deviation (About 68% of data falls within 1 SD, 95% within 2 SD, & 99.7% within 3 SD)

Applications:

- Regression models

- Feature scaling (standardization)

- Improved performance of algorithms like SVM, KNN & Neural Networks

- Bayesian inference & probabilistic models

- Fraud detection, network security

- Used in Gaussian Mixture Models (GMM) for clustering

T-distribution -

Applications:

- Testing a population mean

- Comparing 2 means

- Testing correlation

- Testing for hypothesis of difference between 2 means

Limitations:

- Assumes Normality

- Less useful for large samples

- Sensitive to outliers

- Requires random sampling

RSE (Residual Sum of Squares Error) -

- SD of the residuals (diff between obs values & pred. values)

- Provides an absolute measure of the typical distance that the data points fall from regression line

- Unit: same as dependent variable

- Interpretation: A smaller RSE indicates a better fitting model

$$RSE = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}$$